

# **Econometrics I**

## Lecture 8b: Event Studies

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## Two things called “event study”

- ▶ **The panel event study** (economics, marketing, strategy): some units experience an event — a policy, a product launch, an acquisition — at a known date. Trace the outcome *before and after*, relative to units that did not.
- ▶ **The returns event study** (finance, accounting): a firm announces something. Measure the stock's *abnormal* return around the announcement, relative to what the market model predicts.
- ▶ Same skeleton in both: re-index time so the event is at zero, build a counterfactual path from the pre-period, and read off the gap. What differs is where the counterfactual comes from.
- ▶ Plan: the panel version (with a preview of Week 9's trouble), then the returns version.
- ▶ Attribution: the questions that organize the first half — who is the control, what is normalized, what does a flat pre-period buy you — follow Goldsmith-Pinkham's Applied Methods lecture 14. The examples and the single-timing focus are ours.

## **Panel event studies**

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## Setup: event time

- ▶ Units  $i$ , calendar periods  $t$ . Treated units all experience the event at the same date  $E$ ; the rest never do. (Staggered dates: next week.)
- ▶ **Event time**  $k = t - E$ . The period just before the event is  $k = -1$ .
- ▶ Potential outcomes  $Y_{it}(0), Y_{it}(1)$ . The object we want is a whole *path*:

$$\tau_k = \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid t - E = k, D_i = 1], \quad k = 0, 1, 2, \dots$$

the dynamic average treatment effect on the treated,  $k$  periods after the event.

- ▶ Why a path rather than a number? Effects ramp up (adoption takes time), fade (novelty), or reverse (intertemporal substitution). One coefficient hides all of that, and averages it with weights you did not choose.

$$y_{it} = \alpha_i + \gamma_t + \sum_{k \neq -1} \tau_k \cdot \mathbb{1}\{t - E = k\} \cdot D_i + \varepsilon_{it}$$

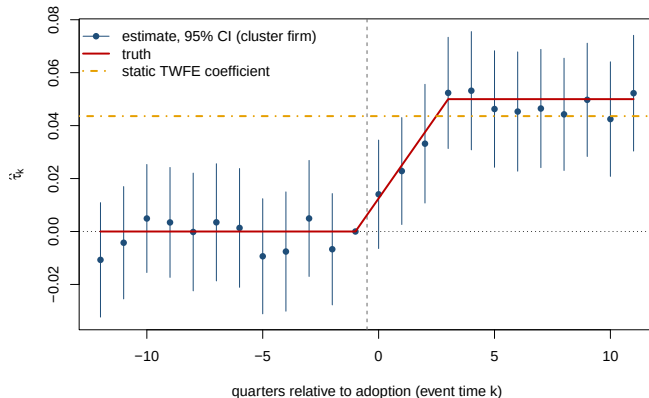
- ▶ Two-way fixed effects (last hour) plus one dummy per event time, switched on only for treated units.
- ▶ **Reference period  $k = -1$  is omitted.** Every  $\tau_k$  is measured relative to the period before the event;  $\hat{\tau}_{-1} \equiv 0$  by construction.
- ▶ Never-treated units have no event-time dummies. They pin down  $\gamma_t$ : what would have happened to everyone absent the event.
- ▶ Pre-period coefficients ( $k < -1$ ) are the *diagnostic*: if the treated group was already diverging, they will show it. Cluster by unit.

## Running example: an algorithmic-pricing tool

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- ▶ 400 firms observed for 24 quarters. In quarter 13, half of them switch on a pricing-optimization tool. The outcome is log gross margin.
- ▶ Ground truth: the effect ramps from 0.0125 in the adoption quarter to 0.05 after three quarters and stays there. Nothing happens before adoption.
- ▶ Also in the data: firm effects  $\alpha_i$ , a common macro path  $\gamma_t$  (random walk with drift), and noise. Treated firms could differ in levels (the firm effects absorb that); in this first version they are on the same trend as the others.
- ▶ Estimate the event-study regression with `fixest's i()`, cluster by firm, and plot.

## The event-study plot



Flat and near zero before, ramping after, sitting on the truth. The dot-dashed orange line is the *single* post-period coefficient from a two-way FE regression with one treatment dummy.

## How to read it

- ▶ The dot at  $k = -1$  is zero *by construction*. Do not read it as evidence.
- ▶ The pre-period dots test “were the treated already moving relative to controls?” Here they are not, and the confidence intervals say how big a divergence we could have seen.
- ▶ The post-period dots are  $\hat{\tau}_k$ : the effect  $k$  quarters after adoption, relative to  $k = -1$ .
- ▶ The static two-way FE coefficient is the average post-period  $\hat{\tau}_k$  minus the average pre-period  $\hat{\tau}_k$  (with  $\hat{\tau}_{-1} = 0$ ); absent pre-trends, an equally weighted average of the  $\tau_k$ ,  $k \geq 0$ . With a ramp, it understates the long-run effect. With effects that fade, it overstates it. It is never the number you would have asked for.
- ▶ Confidence bands are pointwise. Eleven pre-period coefficients at 95% each is not a 95% test of “all flat.” Use a joint (Wald) test, or a uniform band.

## Assumptions, stated properly

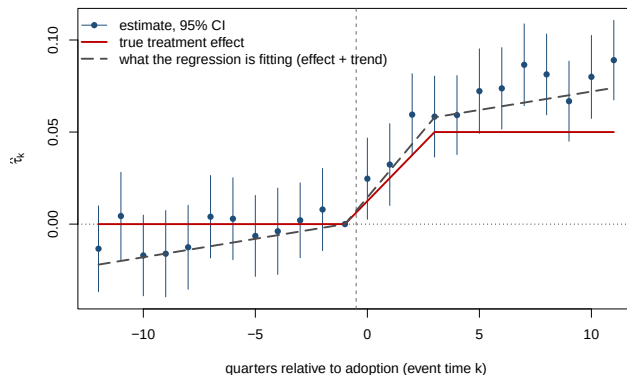
1. **Parallel trends** in event time: absent the event, treated and control paths would have moved together,

$$\mathbb{E}[Y_{it}(0) - Y_{is}(0) \mid D_i = 1] = \mathbb{E}[Y_{it}(0) - Y_{is}(0) \mid D_i = 0] \quad \forall t, s.$$

2. **No anticipation**:  $Y_{it}(1) = Y_{it}(0)$  for  $t < E$ . Units do not act on the event before it happens.

- ▶ Under both,  $\hat{\tau}_k$  is consistent for the dynamic ATT and  $\tau_k = 0$  for  $k < 0$ .
- ▶ Only the *implication* ( $\hat{\tau}_k \approx 0$  for  $k < -1$ ) is testable. Parallel trends in the post-period is an assumption about a counterfactual you never see.
- ▶ Parallel trends is an assumption about *changes* in  $Y(0)$ , not about levels of  $Y$ . Treated firms can be bigger, older, richer. They cannot be on a different trajectory.

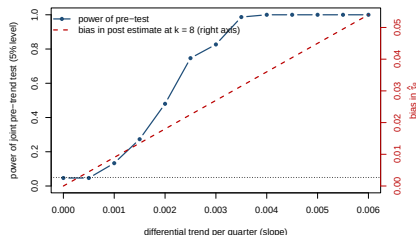
## What a flat pre-period does not buy you



Same design, plus a differential trend of 0.2% per quarter for treated firms. No pre-period dot is individually significant and the joint test does not reject ( $p = 0.30$ ). The post-period estimates are biased upward, and the bias grows with  $k$ .

## The pre-test problem (Roth 2022)

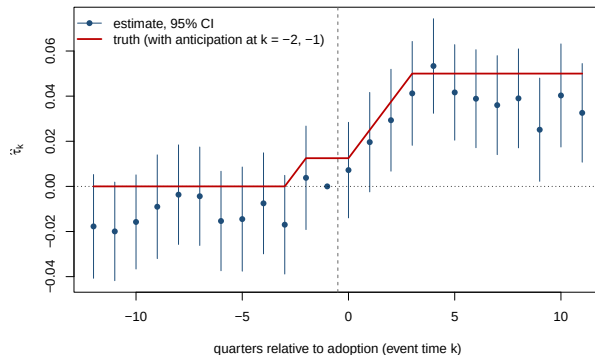
- ▶ A pre-trend test has power only against violations large relative to the noise. When the violation is small, the test passes and the bias is still there.
- ▶ Worse: **conditioning on passing** selects samples where pre-period noise offset the trend; the post-period estimates in those samples are biased *more*, not less.
- ▶ A linear trend with slope  $\delta$  biases  $\hat{\tau}_k$  by  $\delta(k+1)$ : the further out you look, the worse it gets, while the pre-test's power is fixed.



## What to do instead

- ▶ **Plot, and report magnitudes.** “The pre-period coefficients are individually insignificant” is not a finding. “The data rule out a pre-trend larger than  $x$  per quarter, which would bias  $\hat{\tau}_8$  by at most  $9x$ ” is.
- ▶ **Sensitivity analysis.** Rambachan and Roth (2023): allow the post-period violation to be as large as (some multiple of) the largest pre-period violation, and report the range of  $\tau_k$  consistent with that. HonestDiD does it in a few lines.
- ▶ **Longer pre-periods** help detect trends; they do not remove the bias once you find one.
- ▶ **Detrend only with care.** Fitting a treated-group trend on the pre-period and extrapolating it assumes the trend would have continued — a strong claim about a counterfactual, and it eats precision.
- ▶ **A better control group** is worth more than any of the above. Firms in the same segment, matched on pre-period trajectory, or — in Week 11 — a synthetic control.

# Anticipation



Firms start adjusting two quarters before the tool goes live (pilot programs, contract renegotiation). The pre-period now steps up at  $k = -2$ ; since  $k = -1$  is the reference, every *post*-period estimate is shifted down by the anticipation effect. Fix: set the reference to  $k = -3$ , or redefine the event date as the announcement.

## Design choices you will have to make

- ▶ **Who is the control?** Never-treated units are cleanest. If everyone is treated eventually, not-yet-treated units serve until their own event (staggered: next week).
- ▶ **No control group at all?** With a single event date and everyone treated, event time *is* calendar time. The event-time dummies are collinear with  $\gamma_t$ . `fixest` says “without doubt, your model is misspecified,” and it is right: you would be assuming the counterfactual is flat.
- ▶ **Window and binning.** Estimate  $\tau_k$  for  $k \in [-K_0, K_1]$  and bin everything outside into endpoint dummies, so far-out observations do not form a category of three firms.
- ▶ **Balance in event time.** If the panel is unbalanced,  $\hat{\tau}_6$  comes from different firms than  $\hat{\tau}_2$ , and composition changes look like dynamics. Restrict to firms observed over the whole window, or say why not.
- ▶ **Reference period.**  $k = -1$  by convention; earlier if anticipation is plausible.

## Where it comes from: difference-in-differences in disguise

- Take two periods,  $t \in \{E - 1, E + k\}$ , and the two groups. The event-study coefficient is

$$\hat{\tau}_k = (\bar{y}_{E+k}^T - \bar{y}_{E-1}^T) - (\bar{y}_{E+k}^C - \bar{y}_{E-1}^C),$$

a  $2 \times 2$  difference-in-differences comparing period  $k$  to the reference period.

- So the event study is a *collection* of DiD estimates, one per  $k$ , sharing a baseline. Everything you learn about DiD next week applies coefficient by coefficient.
- Equivalently: the regression is fitting each period's treated-minus-control gap and subtracting the gap at  $k = -1$ .

```
es[, k := ifelse(treated == 1, t - 13, -1)]           # controls get the reference value
fit <- feols(y ~ i(k, ref = -1) | firm + t, es, cluster = ~firm)
iplot(fit, ref.line = -0.5)                          # the event-study plot
wald(fit, keep = "k::-")                             # joint test on pre-period coefficients
```

```
es["k"] = np.where(es.treated == 1, es.t - 13, -1)
fit = pf.feols("y ~ i(k, ref=-1) | firm + t", data=es, vcov={"CRV1": "firm"})
fit.iplot()
```

- Binning: recode  $k$  to  $\min(\max(k, -K_0), K_1)$  before  $i()$ .
- Never-treated units should carry the reference value, not NA: they must stay in the regression to estimate  $\gamma_t$ .

## Preview of Week 9: staggered adoption breaks this

- ▶ Today every treated firm adopted in quarter 13. In practice adoption is **staggered**: cohorts  $g = 2019, 2020, 2021, \dots$
- ▶ The same two-way FE regression then uses *already-treated* firms as controls for later cohorts. If effects are dynamic, those “controls” are moving, and the comparison is contaminated (Goodman-Bacon 2021; Sun and Abraham 2021).
- ▶ Symptoms: negative weights on some cohorts; pre-period coefficients that are non-zero even when there is no pre-trend; post-period coefficients that mix cohorts’ dynamics.
- ▶ Fixes (next week): estimate cohort-by-event-time effects and aggregate them yourself (Callaway–Sant’Anna, Sun–Abraham).
- ▶ The single-timing case is the one where two-way FE is exactly right. Know what “right” looks like before we break it.

## Returns event studies

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## The returns event study

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- ▶ Ball and Brown (1968), Fama, Fisher, Jensen and Roll (1969): does the stock market react to news, and how fast? Still the workhorse of empirical accounting and corporate finance.
- ▶ Typical events: earnings announcements, M&A announcements, CEO turnover, index inclusion, regulatory actions, product recalls, data breaches.
- ▶ The logic is the panel event study with one change: the counterfactual for firm  $i$  is not other firms, it is **firm  $i$ 's own relationship to the market**, estimated on pre-event data.
- ▶ The panel is long and narrow per event ( $T \approx 250$  trading days per firm) rather than short and wide. So the “fixed effect” is estimated well, and the interesting inference is cross-sectional: averaging across events.
- ▶ Reference: MacKinlay (1997, *JEL*), “Event Studies in Economics and Finance.”

- ▶ Event day 0 for firm  $i$ . **Estimation window**  $[-250, -11]$ ; **event window**  $[-10, +10]$ . Ending the estimation window before the event window keeps the event out of the estimation sample.

- ▶ **Market model**, estimated on the estimation window by OLS, one firm at a time:

$$R_{it} = a_i + b_i R_{mt} + e_{it}, \quad \mathbb{E}[e_{it} \mid R_{mt}] = 0.$$

- ▶ **Abnormal return** on the event window: actual minus predicted,

$$\widehat{AR}_{it} = R_{it} - \hat{a}_i - \hat{b}_i R_{mt}.$$

- ▶ **Cumulative abnormal return** over a window  $[k_1, k_2]$ :  $\widehat{CAR}_i(k_1, k_2) = \sum_{t=k_1}^{k_2} \widehat{AR}_{it}$ .
- ▶ Compare to the panel version:  $a_i$  is  $\alpha_i$ ,  $b_i R_{mt}$  is  $\gamma_t$  with a firm-specific loading, and  $AR_{it}$  is  $\hat{\tau}_k$  for one firm.

## Simulated example: earnings announcements

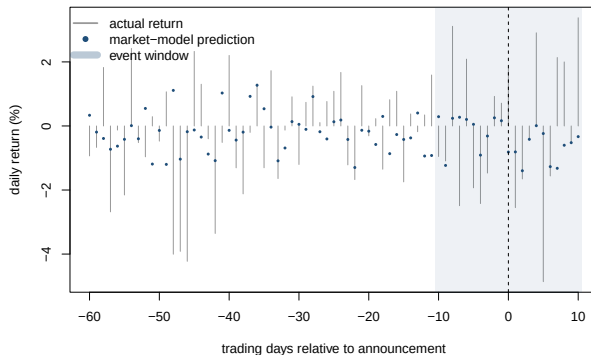
- ▶ 200 firms, each with one announcement. Daily returns for 261 days per firm.
- ▶ Market return  $R_{mt} \sim N(0.04\%, 1\%^2)$ , common to all firms on a given day. Betas  $b_i \sim N(1, 0.3^2)$ ; idiosyncratic volatility 2% per day.
- ▶ Each firm has a standardized **earnings surprise**  $S_i \sim N(0, 1)$ . Ground truth for the abnormal return:

$$AR_{i0} = 0.02 S_i, \quad AR_{it} = 0.001 S_i \text{ for } t = 1, \dots, 10, \quad 0 \text{ otherwise.}$$

A one-SD surprise moves the stock 2% on the day and drifts another 1% over two weeks (post-earnings-announcement drift, Bernard and Thomas 1989).

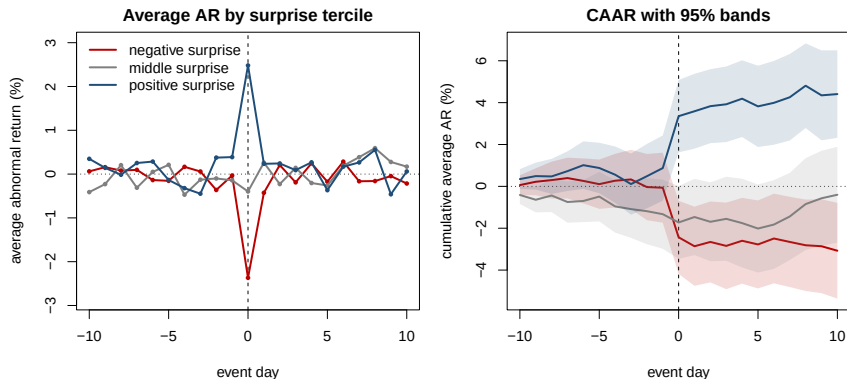
- ▶ So  $CAR_i(0, 10) = 0.03 S_i$  in expectation, and  $CAR_i(-10, -2) = 0$ .

## One firm



The firm with the largest positive surprise. Grey bars are actual daily returns; blue dots the market-model prediction. The abnormal return is the gap. Even for the largest surprise, a true 5.9% jump shows up as a 2.6% abnormal return after a bad noise draw, smaller than the move on day +5. One firm tells you little.

## Averaging across firms



Left: the average abnormal return by event day, by tercile of surprise. Right: cumulated. The day-0 jump and the post-announcement drift are both visible once you average over  $\approx 67$  firms. Nothing happens before day 0: the analog of a flat pre-period.

- ▶ Under the null of no abnormal return,  $\widehat{AR}_{it}$  has mean zero and variance  $\sigma_{e,i}^2 (1 + \text{small estimation-error term})$ ; over a window of length  $L$ ,  $\mathbb{V}[\widehat{CAR}_i] \approx L \sigma_{e,i}^2$ .
- ▶ Two standard test statistics for the **average** CAR across  $N$  events:
  - **Cross-sectional  $t$** : treat the  $N$  CARs as an i.i.d. sample. Robust to event-induced variance changes; needs  $N$  moderately large.
  - **Standardized** (Patell 1976): scale each firm's CAR by its own  $\hat{\sigma}_{e,i} \sqrt{L}$  first, then average. Better power when volatility differs across firms.
- ▶ Both assume independence *across firms*. That fails when events cluster in calendar time (every firm announces in the same week; a regulation hits every bank on the same day). Then the market-model residuals are cross-sectionally correlated and the  $t$ -statistics are too large — the same clustering problem as Week 3, in a different costume. Fix: calendar-time portfolios, or cluster by event date.

## CARs by window

| Window      | CAAR (%) | $t$ (cross-section) | $t$ (standardized) | CAAR, $S > 0$ | CAAR, $S < 0$ |
|-------------|----------|---------------------|--------------------|---------------|---------------|
| $[-10, -2]$ | -0.25    | -0.54               | -0.57              | 0.17          | -0.56         |
| $[-1, 1]$   | -0.02    | -0.05               | -0.08              | 2.71          | -2.03         |
| $[0, 0]$    | -0.11    | -0.53               | -0.75              | 2.06          | -1.71         |
| $[1, 10]$   | 0.58     | 1.38                | 1.30               | 1.22          | 0.10          |
| $[0, 10]$   | 0.47     | 1.01                | 1.02               | 3.28          | -1.61         |

- The pre-event window is indistinguishable from zero; the announcement windows are not, once you split by the sign of the surprise. (Pooled over all firms, the average CAR is near zero because surprises average to zero. The *sign* of  $S_i$  is the treatment.)
- The two test statistics agree here because all firms have similar volatility.

## The cross-sectional regression

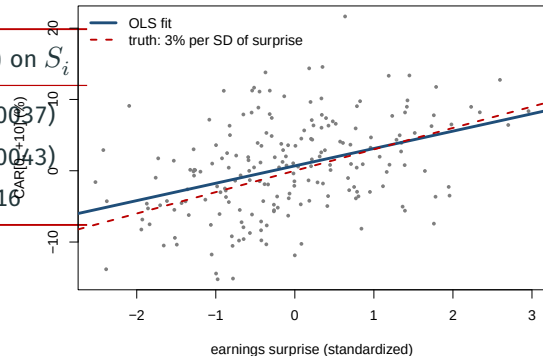
The quantity accounting cares about is the **earnings response coefficient**: how much return per unit of surprise.

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|                                         |                                   |
|-----------------------------------------|-----------------------------------|
|                                         | $\widehat{CAR}_i(0, 10)$ on $S_i$ |
| Slope on surprise $S_i$ (truth = 0.030) | 0.0244 (0.0037)                   |
| Intercept                               | 0.0068 (0.0043)                   |
| $N$ firms, $R^2$                        | 200, 0.16                         |

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Regress  $\widehat{CAR}_i(0, 10)$  on  $S_i$  with heteroskedasticity-robust standard errors. The slope is within sampling error of the truth: the CAR analog of interacting event-time dummies with a covariate.



## What can go wrong

- ▶ **Confounding events.** A dividend change announced with earnings; a merger rumor the day before. Short windows help; reading the news for each event helps more.
- ▶ **Event-date uncertainty.** Information leaks; after-hours announcements land on the next trading day. Widen the window, at the cost of noise.
- ▶ **The joint-hypothesis problem.** “Abnormal” is relative to a model. If the market model is wrong (missing factors, time-varying beta), so is the abnormal return. Over days it barely matters; over months or years (long-horizon “BHAR” studies) it dominates.
- ▶ **Thin trading and non-synchronous prices** bias  $\hat{b}_i$  for small stocks.
- ▶ **Selection into the event.** Firms choose when to announce, whether to do a buyback, whether to be acquired. The CAR measures the market’s reaction *conditional on the firm having chosen the event*. That is often what you want; it is not the effect of assigning the event at random.

## Recap

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- ▶ An event study re-indexes time to the event and compares the outcome path to a counterfactual path.
- ▶ **Panel version**: two-way FE plus event-time dummies, reference  $k = -1$ , never-treated controls estimate  $\gamma_t$ . Needs parallel trends and no anticipation. The pre-period is a diagnostic with limited power; report what it rules out, not just whether it passed.
- ▶ **Returns version**: the market model is the counterfactual, estimated per firm on the pre-window. Abnormal returns, cumulated and averaged; cross-sectional  $t$  or standardized tests; watch for clustered event dates.
- ▶ Both are collections of difference-in-differences. Next week: what happens when the events happen at different times.

`inclass_session8.Rmd`, second half:

1. Reproduce the event-study plot and the joint pre-trend test. Then add anticipation and fix it by changing the reference period.
2. Reproduce the CAR windows. Then move every firm's announcement to the same calendar day and add a common market shock on that day. What happens to the  $t$ -statistics, and why?